

BAO NGUYEN

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Relevant Experience

M2M Tech

Surrey, BC

Machine Learning Engineer Intern

Apr 2024 - Present

- Proposed and implemented an agent orchestration architecture for a digital twin application
- Implemented an agentic pipeline to automate end-to-end ML development lifecycle, FastAPI and MCP integration for seamless inference and endpoint registration, reducing development lead time by 40%
- Created a full Python data processing pipeline for model training by cleaning a dataset of over 200k records and engineering features from Nemoh, an open source BEM software
- Developed an end-to-end integration of digital twin models using **Azure** (Event Hub, Data Explorer, Function App, Digital Twins, AutoML) and Power BI to integrate real-time data analysis and optimization using machine learning
- Designed and taught curriculums for Python, Power BI and Azure to over 40 students
- Compiled and trained a dataset of 100 articles using **LangChain** and self-critique prompting to extract domain-specific entity information, then configured and presented results using **LangSmith**
- Implemented a data pipeline with **LLM classification** for 790 prompts to extract high-quality prompts for **web scraping**, display Google and Bing scraping results, and determine prompt quality using **ROUGE** and other metrics
- Transitioned code into production by integrating **pytest**, providing technical and functional documentation, as well as organizing an existing codebase into the desired code architecture to satisfy project requirements

NXT-GEN, University of Alberta

Edmonton, AB

Vice President Marketing

June 2022 - Aug 2024

- Led a marketing team of 5 students to promote events, workshops, and competitions as well as maintain active presence on Instagram, LinkedIn and club website
- Reached out to 20 high schools and recruited over 50 students into for the Accelerator Program, an entrepreneurship initiative for high school and university students with over \$6,000 of competition prizes
- Created Canva posts to promote events, competitions and other organization activities

Sponsorships Associate

Oct 2021 - May 2022

- Conducted outreach to over ten companies for sponsorship and collaboration opportunities
- Successfully initiated partnership with local KPMG branch to reach an \$800 sponsorship deal and Junior Achievement to recruit over 10 new members

Wayfinders Business Co-operative

Edmonton, AB

Project Budget Planner

Dec 2023 - Feb 2024

- Prepared Excel financial budgets for entrepreneurial events and projects, satisfying all internal requirements and reducing costs by 20 percent
- Revised the company's general budget, resulting in a 10% cost decrease after updating current revenue streams and optimizing expenses
- Compiled a list of grants and loans worth over \$100,000 to help raise funds for the above projects

Youreka Canada

Edmonton, AB

Undergraduate Research Lead

Feb 2023 - Apr 2023

- Wrote a 10-page research project with 3 high school students on mental health services for Canadian youth over the COVID-19 pandemic
- Prepared and analyzed government data using R libraries
- Compiled and presented findings at an annual research symposium of over 50 attendees

Active Learning

Edmonton, AB

Private Tutor

Sep 2021 - Sep 2023

- Tutored over 10 students from elementary to university levels for a combined total of over 40 students tutored throughout 4 years of tutoring experience
- Regularly updated parents about students' progress for in-person and online tutoring
- Gained over 5 students from advertising company on Kijiji, Superprof and other online platforms

Education

University of Alberta

Edmonton, AB

Bachelors of Science Specialization in Mathematics and Finance

Sep 2021 - Present

- **Expected Graduation:** June 2026

Personal Projects

Canadian Rental Price Prediction

Oct 2023 - Present

- Created a mathematical model based on the Markov voter model to predict average rental prices of cities in Canada
- Performed maximum likelihood estimation in Python

Standard Metropolitan Statistical Area

Oct 2023 - Present

- Predicted serious crime rate using a linear regression model with 11 independent variables in R
- Compiled a 15-page report about the case study, methodology, and results

Technical Skills

Languages: Python, R

Technologies: NumPy, Pandas, TensorFlow, Scikit-learn, car, dplyr, tidyverse, ggsignif, Git, LaTeX, Linux

Achievements

Second Place in NXT-GEN Business Pitch Competition

May 2021

Grades 10, 11 and 12 Honours with Distinction

2018 - 2021

Top 50 students in Alberta High School Math Contest First Round

2019

Performance with Honours in the Canadian Open Math Contest

2018, 2019